INFO 4555 / 5556: Business Intelligence Systems

Assignment 1: Data Model for Grocery Store Point of Sale

Background:

When gathering requirements for a business process, you may not initially have access to the source system. However, you may have some indirect access, which could be used as indicators to the data attributes available in the source system.

Over the summer Jeff decided to invite some friends over for hamburgers and beer. On his way home from work, Jeff stopped by the P&C grocery store and purchased the following items:

- Ground beef patties (pre-packaged hamburger patties)
- Sandwich buns (hamburger buns)
- Orange Juice
- Fat Tire 6pack (good beer from Colorado)
- Bags of ice (keep the beer cold)

With the receipt from Jeff's purchases from P&C, you can derive some basic information about the P&C Point of Sales system.

Assignment:

Using the provided sales receipt, create a dimensional data model that would support basic sales analysis for the P&C grocery store chain. Create a logical data model, along with the assumptions you made, and questions you come up with for the P&C executives requesting a BI solution to support sales analysis.

Please provide the following deliverables:

- Data Model containing dimension tables and a fact table based on the data attributes found on the receipt.
    - should include attribute names (you can make them up, but make them meaningful)
    - should include any primary, foreign, natural keys, etc, and label them as seen in text book and lecture slides.
    - apply dimensional model concepts discussed in lectures and reading
    - Include attributes you feel should be included, but may not exist on the receipt. These attributes should be included in your assumptions.
- A document stating the business process modeled, grain of the fact table, any assumptions you made while creating your model (at least 3), and questions for the P&C executives (at least 3).
    - Include a list of questions you would have for the Business customer. Questions should be requirement clarifying questions.

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Turning your assignment in:

- Should turn in a single PDF file
    - Times New Roman font, heading size 14, content size 12, single spaced
- You may use any drawing tool for the conceptual model (example. Visio, Powerpoint, Word) make sure you save as PDF before uploading to Canvas.
- A hand-drawn data model is acceptable, as long as, it is neat, legible, and scanned to PDF format.
- Assignment must be turned in via Canvas and by assignment deadline.

Grading Criteria:

- All assigned deliverables included
- Assignment turned in on time
- The Data Model and supporting documents include the concepts discussed in class readings, lectures, and discussions, chapter 3.
- The dimensional model may be hand drawn and scanned to PDF, but it should be legible and use standard found in book and lecture slides.
    - Should include table names and column names
    - Should include primary and foreign keys with (PK) or (FK) table
    - Should include lines to indicate the relationships between the entities (tables)
    - Should include any other notable items found in lecture slides/text and identified in receipt
- Makes a few assumptions when creating your data model based on chapter 3, and be sure to list them under your assumptions. See below.
- Statement of your fact table grain.
- At least 3 assumptions made in data model, such as attributes not found on receipt, based on examples in book and class.
- At least 3 questions for the Business person in charge of the grocery store POC system.

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P&C FRESH #3107
315 PINE TREE RD
ITHACA NY 14850
(607)272--5836

		HOLLY
3107 01 01392527 07/23/16	5:30PM	117
PF SANDWICH BUNS	$3.49	F
PF SANDWICH BUNS	$3.49	F
BYRNE ORG JUICE	$1.79	F
GRND CHUCK PATTY	$4.84	F
GRND CHUCK PATTY	$4.94	F
GRND CHUCK PATTY	$4.94	F
GRND CHUCK PATTY	$4.99	F
BLGM FAT TIRE 6P	$9.99	T
BOTTLE DEPOSIT	$0.30	
STRICKLER BAG IC	$1.89	TF
STRICKLER BAG IC	$1.89	TF
SUBTOTAL	$42.55	
SALES TAX	$1.10	
TOTAL	$43.65	
VISA	$43.65	

ACCT# XXXXXXXXXXXX[redacted]
JEFFREY R CHRISTEN
07/23/16 17:32
AUTH # 023898
REF # 016557
SEQ # 016557
MID # 923776

CHANGE	$0.00

CUSTOMER BORN: [redacted]
OF ITEMS: 10

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